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# Introduction to Economics

Class 3 — The frontier, comparative advantage, the circular flow

**Pavel Kocourek**

Universitat de Barcelona · Group A6

28 September 2026

## Plan for today

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|       |                                          |
|-------|------------------------------------------|
| 10:00 | Warm-up, in pairs                        |
| 10:08 | The production possibilities frontier    |
| 10:32 | Two people, two goods: who makes what    |
| 11:00 | Break                                    |
| 11:10 | Comparative advantage, between countries |
| 11:25 | The circular flow                        |
| 11:41 | One more problem, and Block 1 so far     |
| 11:52 | Before Thursday                          |

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## Warm-up — four more tables

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**A workshop has ten workers. In a day, each can make 4 chairs or 2 tables. Today six make chairs and four make tables. The owner wants four more tables. What does that cost?**

- A)** Four chairs
- B)** Eight chairs
- C)** Two workers' wages
- D)** Nothing — the workers are paid anyway

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Four tables need two workers; moved from chairs, they stop making eight a day. Wages are paid either way — not the cost of the decision.

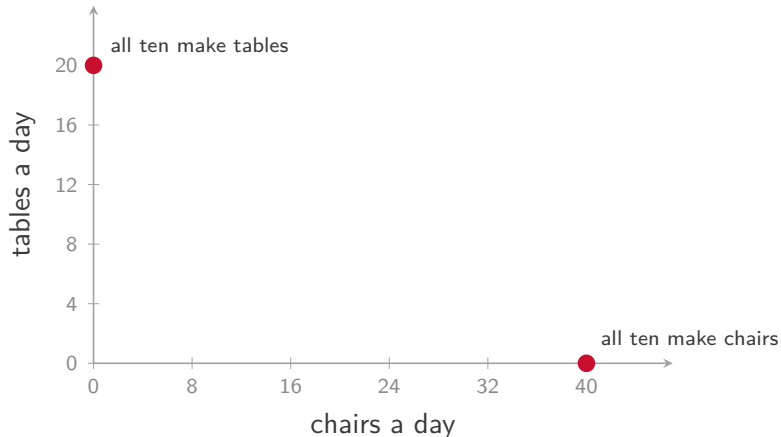
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# The production possibilities frontier

The warm-up, drawn

## Every point on the line uses all ten workers

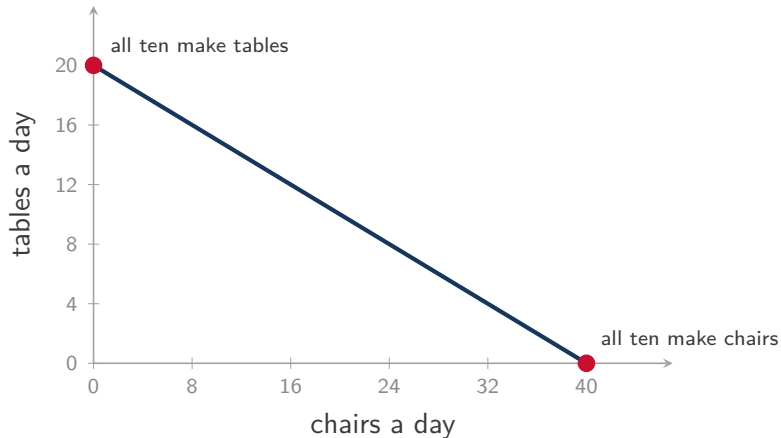
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The **frontier**: the most of one good you can make, given how much of the other you make.

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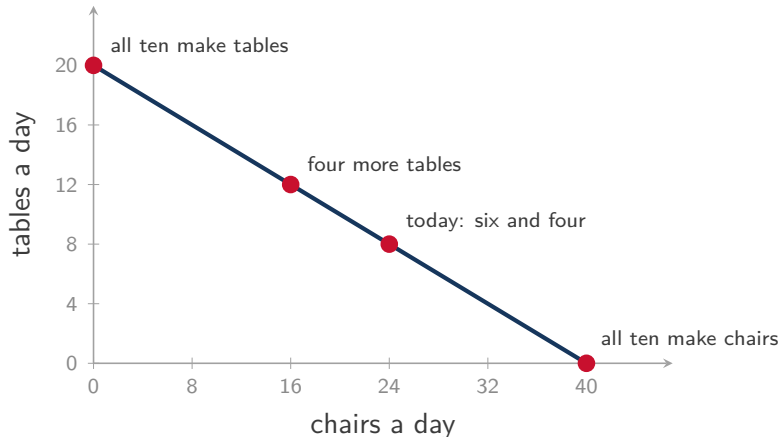
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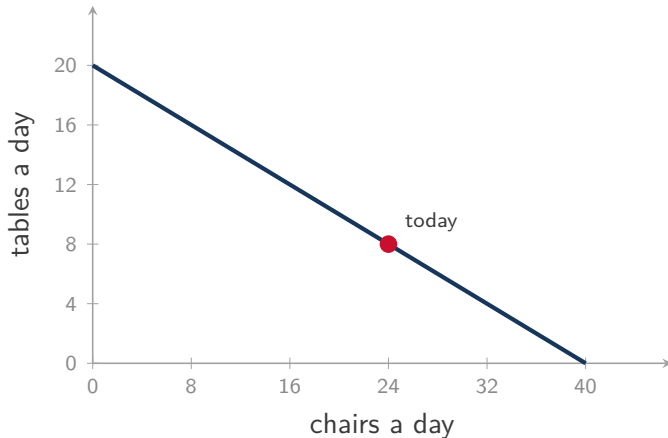
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The **frontier**: the most of one good you can make, given how much of the other you make.

## Inside: idle workers. Outside: not enough of them

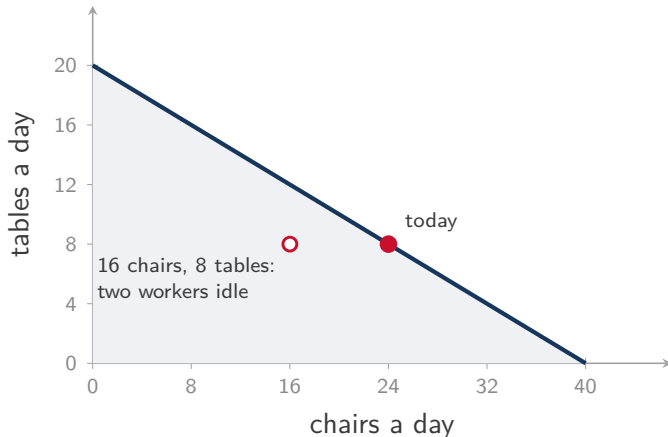
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On the line: **efficient**. Inside: possible, wasteful. Outside: not with these resources.

## Inside: idle workers. Outside: not enough of them

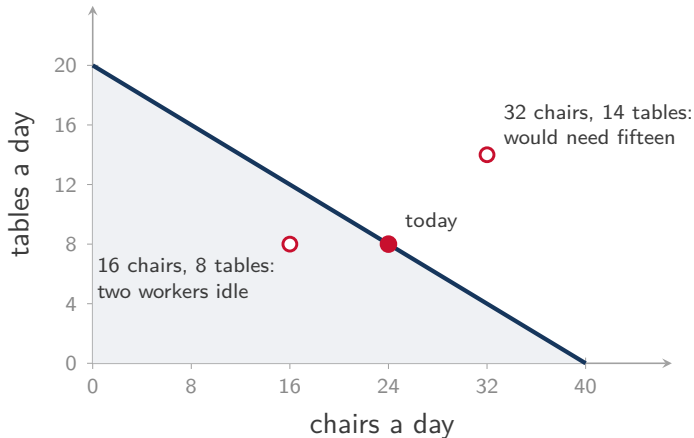
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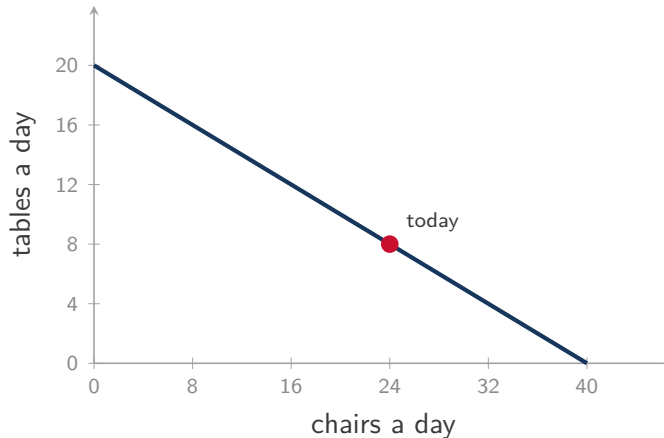
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On the line: **efficient**. Inside: possible, wasteful. Outside: not with these resources.

## The slope is the price: one table costs two chairs

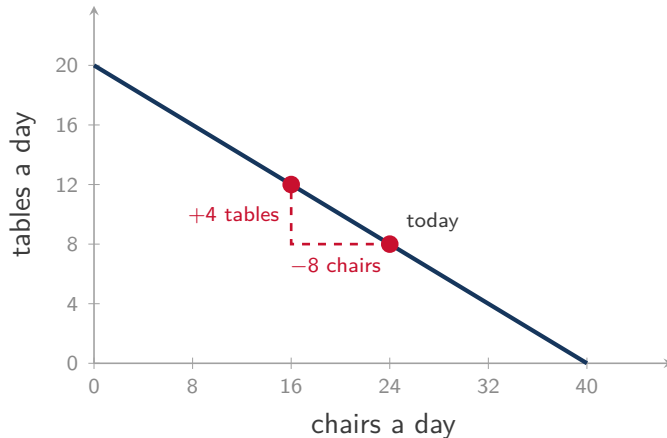
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**Opportunity cost** of a table = chairs given up  $\div$  tables gained — 2, everywhere on the line.

## The slope is the price: one table costs two chairs

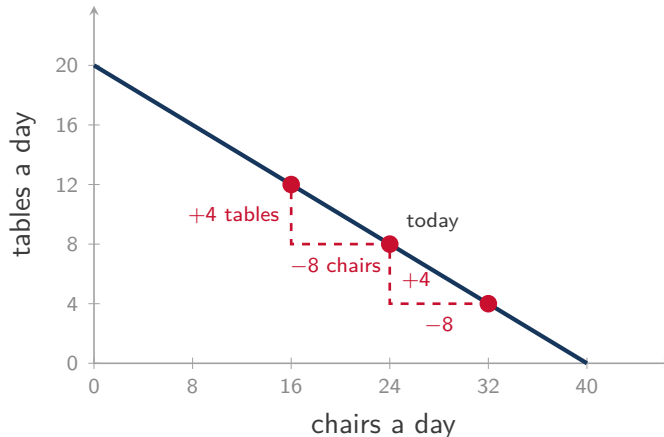
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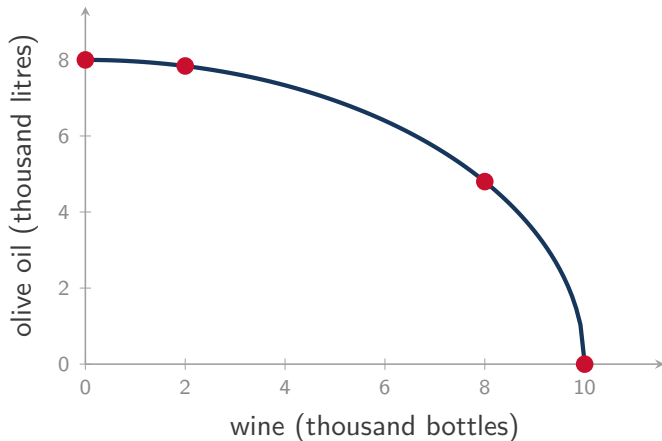
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## Real frontiers bow out: the last field costs the most

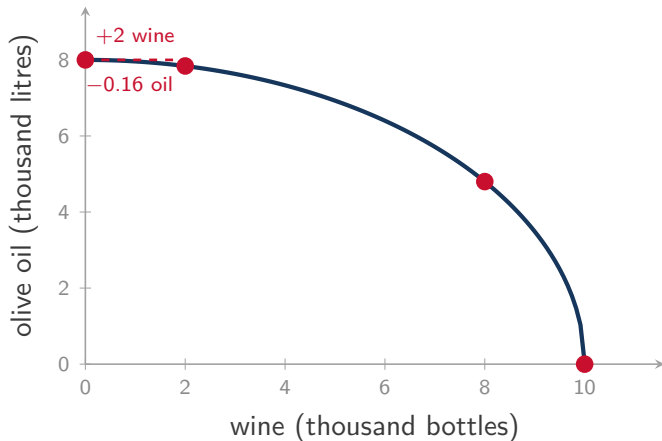
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Fields differ: the first moved to vines were poor olive land; the last were the best.

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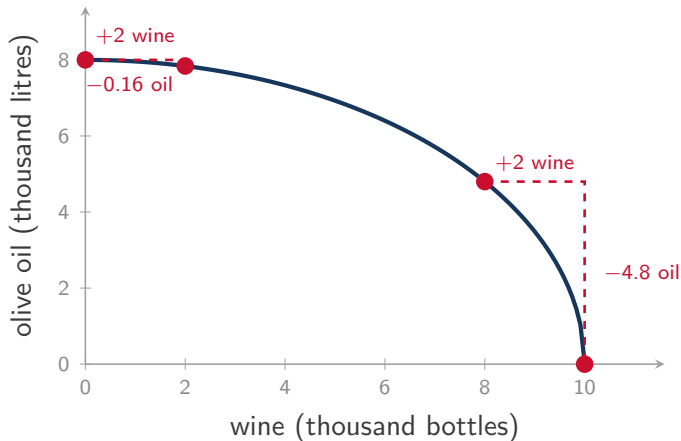
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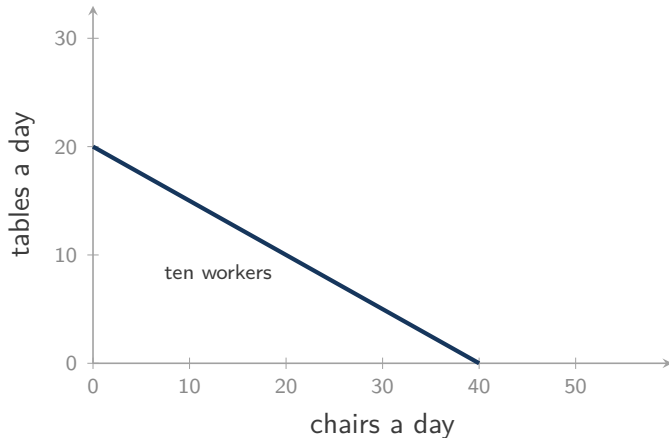
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## More workers push the line out; a better tool moves one end

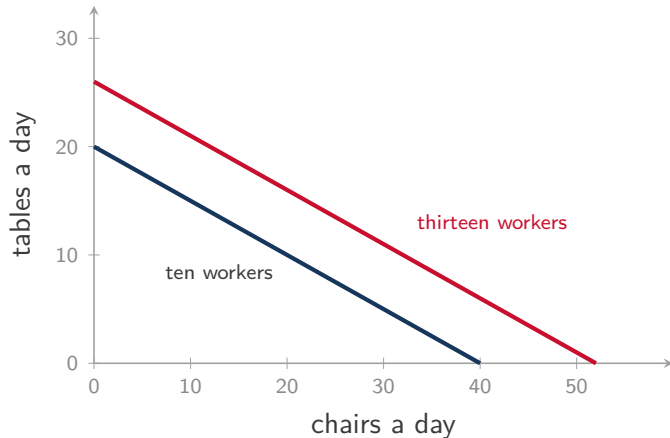
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**Growth** is the frontier moving out: more resources, or better technology.

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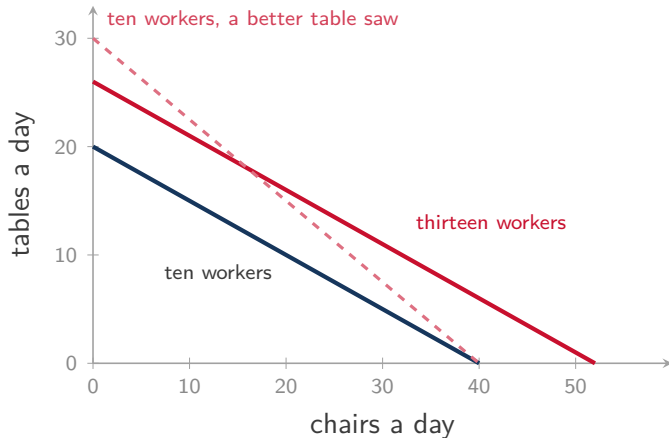
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# Specialisation and trade

Two people, two goods, one stall

## Laia is faster at both. Should she make everything?

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| in one hour | coffees | bocadillos |
|-------------|---------|------------|
| <b>Laia</b> | 12      | 4          |
| <b>Marc</b> | 3       | 3          |

Four-hour shifts. Today each spends two hours on each: Laia 24 coffees and 8 bocadillos, Marc 6 and 6.

## Who should make the bocadillos?

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**Same four hours each, but now they can split the work and swap. Who should spend more time on bocadillos?**

- A)** Laia — she is faster at both
- B)** Marc: one coffee per bocadillo; Laia: three
- C)** Half each, to be fair
- D)** It makes no difference

## Who should make the bocadillos?

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Same four hours each, but now they can split the work and swap. Who should spend more time on bocadillos?

A) Laia — she is faster at both

**B) Marc: one coffee per bocadillo; Laia: three ✓**

C) Half each, to be fair

D) It makes no difference

An hour on bocadillos costs Laia 12 coffees for 4 — three each; Marc 3 coffees for 3 — one each.

## Compare what each gives up, not how fast each is

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| opportunity cost of | one coffee              | one bocadillo |
|---------------------|-------------------------|---------------|
| Laia                | $\frac{1}{3}$ bocadillo | 3 coffees     |
| Marc                | 1 bocadillo             | 1 coffee      |

**Absolute advantage:** Laia, both goods. **Comparative advantage:** Laia coffee, Marc bocadillos.

## Five bocadillos for ten coffees: both end up with more

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|       | Laia                  |                      | Marc                 |                      |
|-------|-----------------------|----------------------|----------------------|----------------------|
|       | coffees               | bocadillos           | coffees              | bocadillos           |
| Today | 24 <small>2 h</small> | 8 <small>2 h</small> | 6 <small>2 h</small> | 6 <small>2 h</small> |
|       |                       |                      |                      |                      |
|       |                       |                      |                      |                      |

The price, 2 coffees a bocadillo, sits between Marc's cost (1) and Laia's (3) — so both say yes.

## Five bocadillos for ten coffees: both end up with more

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|             | Laia                  |                      | Marc                 |                       |
|-------------|-----------------------|----------------------|----------------------|-----------------------|
|             | coffees               | bocadillos           | coffees              | bocadillos            |
| Today       | 24 <small>2 h</small> | 8 <small>2 h</small> | 6 <small>2 h</small> | 6 <small>2 h</small>  |
| Hours moved | 36 <small>3 h</small> | 4 <small>1 h</small> | 0 <small>0 h</small> | 12 <small>4 h</small> |

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| Swap              | -10                   | +5                   | +10                  | -5                    |
| Each ends up with | <b>26</b>             | <b>9</b>             | <b>10</b>            | <b>7</b>              |

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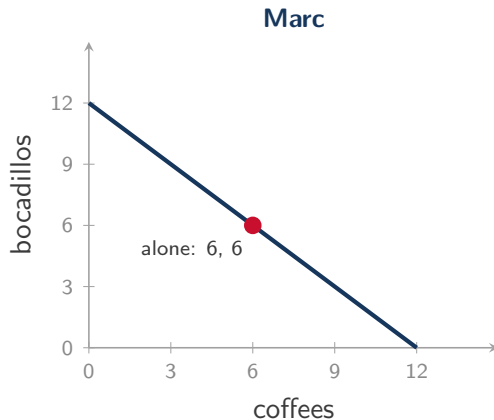
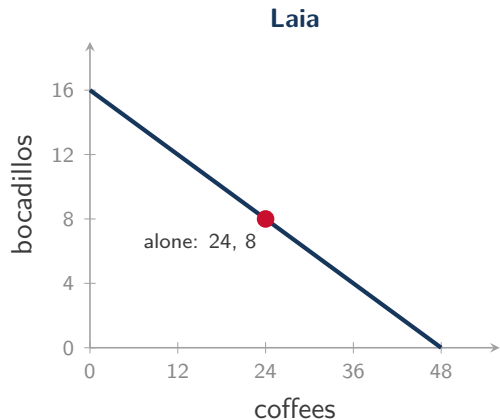
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| Each ends up with | <b>26</b>             | <b>9</b>             | <b>10</b>            | <b>7</b>              |
| Gain              | <b>+2</b>             | <b>+1</b>            | <b>+4</b>            | <b>+1</b>             |

The price, 2 coffees a bocadillo, sits between Marc's cost (1) and Laia's (3) — so both say yes.

## With trade, each eats at a point beyond their own frontier

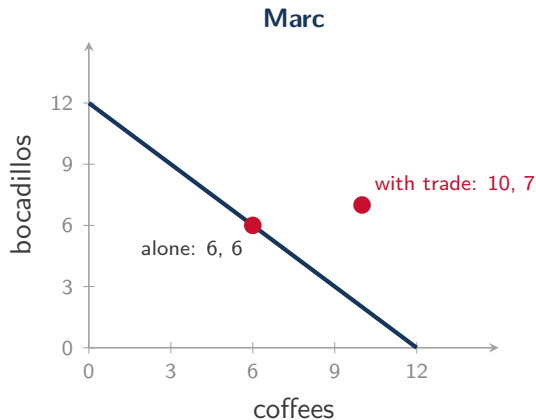
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Production stays on the frontier. **Trade** moves what you can *consume* beyond it.

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# Break

Ten minutes

## Materials, and the second hour

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- Materials: [pkocourek.com/teaching/intro-economics](https://pkocourek.com/teaching/intro-economics)
- Now: comparative advantage between countries, then the circular flow

## Faster is absolute; cheaper to give up is comparative

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### Absolute advantage

- Who needs fewer hours
- Laia, in both goods
- One side can hold it in everything

### Comparative advantage

- Who gives up less
- Laia coffee, Marc bocadillos
- Nobody holds it in everything

Trade follows the second, not the first. Ricardo, 1817 — and Marc's stall.

## Each country sells what costs it least to make

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40 %

of the world's olive oil is pressed in  
Spain

14 %

of the world's machinery exports leave  
Germany

Spain has engineers and Germany has olive groves. Each does *more* of what it gives up least for.

IOC, crop year 2024/25: 1.42 of 3.57 million tonnes · VDMA, machinery export shares 2024.

## The country gains; not every worker does

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- Exporters and their workers gain
- Buyers get cheaper imports
- Workers in the import-competing industry lose
- The gains outweigh the losses — sharing them is politics

Imports: made abroad, bought here. Exports: made here, sold abroad.

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# The circular flow

Households, firms, two markets

## Goods go round one way, money the other

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**Households**

**Firms**

Two loops: things (blue) one way, euros (red) the other. Every sale is someone's income.

## Goods go round one way, money the other

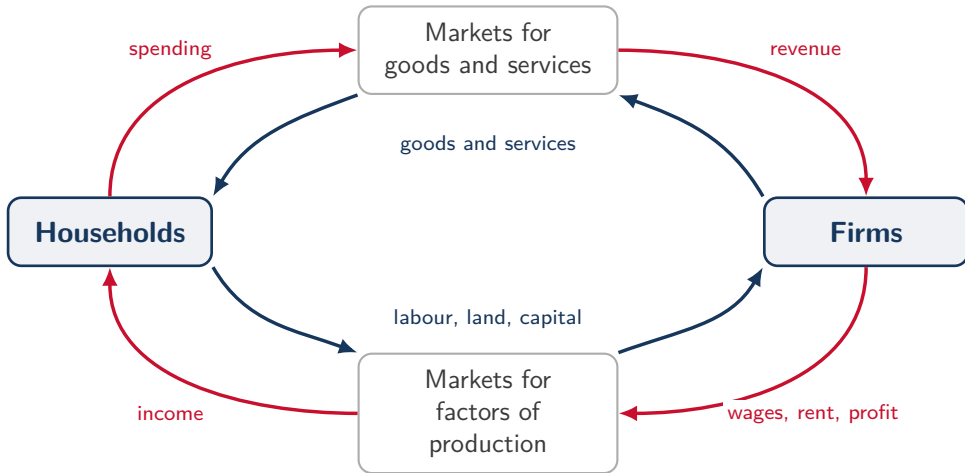
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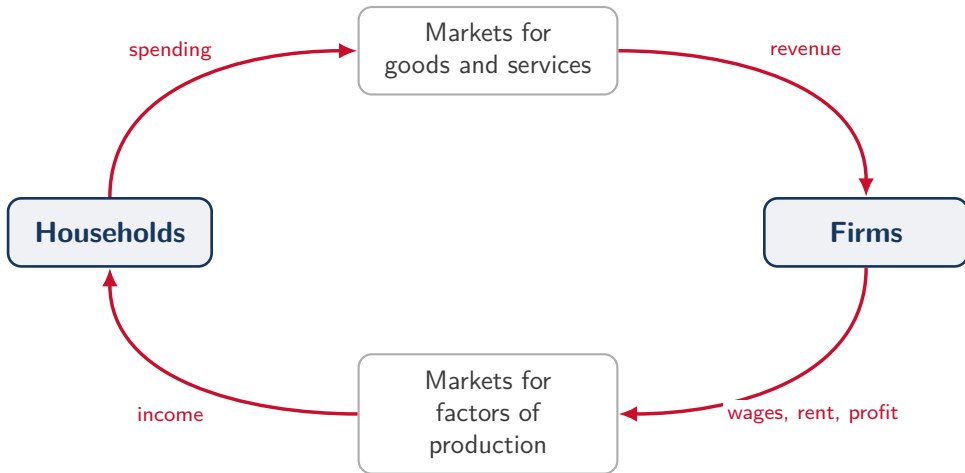
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## Some income leaks out; spending is injected back

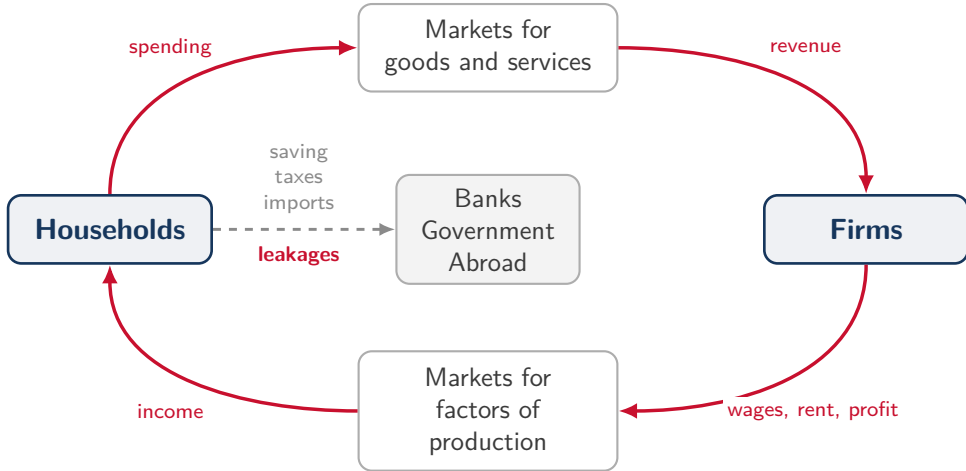
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Out through banks, the state and abroad — and back in through the same three doors.

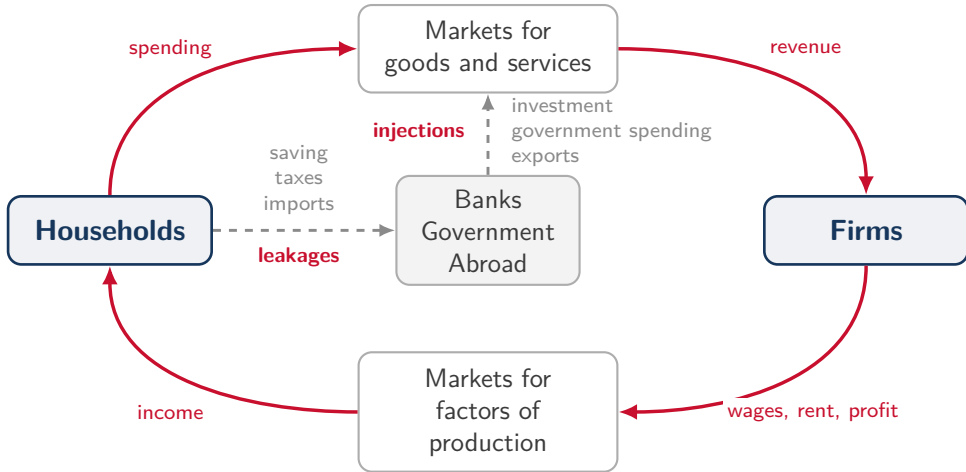
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Out through banks, the state and abroad — and back in through the same three doors.

## Some income leaks out; spending is injected back



Out through banks, the state and abroad — and back in through the same three doors.

Every euro spent is a euro someone earns

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**Spain, 2024: about 1.6 trillion euros**

whichever of the three ways you count it

**GDP:** the same flow measured three ways. Block 5 comes back to it.

INE, Contabilidad Nacional Anual de España, GDP at current prices 2024: 1,594 billion euros.

## Problem — Ricardo's question, with new numbers

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In a day, a Portuguese worker makes 8 barrels of wine or 8 bolts of cloth; an English worker 2 barrels or 4 bolts. Who should make what?

A) Portugal makes both — it is better at both

B) Portugal wine, England cloth

C) England wine, Portugal cloth

D) No trade: England has nothing to offer

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C) England wine, Portugal cloth

D) No trade: England has nothing to offer

A barrel costs Portugal 1 bolt, England 2 bolts: Portugal has the comparative advantage in wine. A bolt costs England half a barrel, Portugal a whole one.

## Block 1 so far, and Thursday's question

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- Scarcity draws the frontier
- Opportunity cost is its slope
- Comparative advantage takes you beyond it
- The circular flow: spending is income

“Spain exports olive oil” and “Spain *should* export more”: which one can be wrong? Thursday.

## Before Thursday 1 October

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- Read Mankiw & Taylor, Ch. 2, pp. 26–29 — positive and normative statements
- Practise today's unit in Sextant: *Trade, the frontier and the flow*
- After class (optional): *Moving frontiers and the price of a deal*
- Thursday 1 October, 10:00 — Block 1 wrap-up
- Thursday: project groups of 2–3 — see the course page



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